

SERIAL COMMUNICATION

ANUSHA N

8051 SERIAL COMMUNICATION

To allow data transfer between the PC and an 8051 system without any error.

- The 8051 supports a full duplex serial port. Full duplex means that it can transmit and receive a byte simultaneously.
- The 8051 has **TXD (pin 11 or P3.1)** and **RXD (pin 10 or P3.0)** pins for transmission and reception of serial data respectively. These pins are TTL compatible.
- The 8051 transfers and receives data serially with different baud rates. Three special function registers support serial communication, namely, SBUF Register, SCON Register and PCON Register.

SPECIAL FUNCTION REGISTERS

- *SBUF Register* SBUF is an 8 bit register. It has separate SBUF registers for data transmission and for data reception. These two registers are accessed by the same name, SBUF.
- One of the registers is write only and used to hold data to be transmitted via TXD pin. The other is read only and holds the received data from external source via RXD pin.
- For a byte of data to be transferred via the TXD line, it must be placed in the SBUF register. Similarly, SBUF holds the 8 bit data received by the RXD pin.
- *SCON Register* This register contains mode selection bits, serial port interrupt bit (TI and RI) and also the ninth data bit for transmission and reception (TB8 and RB8).

SERIAL CONTROL REGISTER (SCON)

Bit 7				Bit 0			
SM0	SM1	SM2	REN	TB8	RB8	TI	RI
Serial communication mode selection bits							
SM0	SM1	Mode	Description	Baud rate			
0	0	Mode 0	8 bit shift register mode	fosc/12			
0	1	Mode 1	8 bit UART	Variable (set by timer 1)			
1	0	Mode 2	9 bit UART	fosc/32 or fosc/164			
1	1	Mode 3	9 bit UART	variable (set by timer 1)			
SM2	In modes 2 and 3, if set, this will enable multiprocessor communication						
REN	Enables serial reception						
TB8	This is the 9 th data bit that is transmitted in modes 2 and 3						
RB8	9 th data bit that is received in modes 2 and 3, it is not used in mode 0 In mode 1, if SM2 = 0, then RB8 is the stop bit that is received						
TI	Transmit interrupt flag, set by hardware must be cleared by software						
RI	Receive interrupt flag, set by hardware must be cleared by software						

Serial Control Register (SCON)

SCON Register

- SCON is an 8-bit register used to program the start bit, stop bit, and data bits of data framing, among other things

	SM0	SM1	SM2	REN	TB8	RB8	TI	RI
SM0	SCON.7							
SM1	SCON.6							
SM2	SCON.5							
REN	SCON.4							
TB8	SCON.3							
RB8	SCON.2							
TI	SCON.1							
RI	SCON.0							

Note: Make SM2, TB8, and RB8 = 0.

Figure 10-9. SCON Serial Port Control Register (Bit-Addressable)

✓ **SCON Serial Control register –**
{Bit Address SCON.7 to SCON.0}

SM0	SM1	SM2	REN	TB8	RB8	TI	RI
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❑ **SM0 & SM1 – Mode Control bits**

SM0	SM1	Serial Mode	Description	Baud Rate
0	0	Mode 0	Shift Register	Fosc/12
0	1	Mode 1	8 bit UART	Variable
1	0	Mode 2	9 bit UART	Fosc/32 or Fosc/64
1	1	Mode 3	9 bit UART	Variable

❑ **Mode 0 {Shift Register sends only data}**

D0	D1	D2	D3	D4	D5	D6	D7
----	----	----	----	----	----	----	----

❑ **Mode 1 {8 Bit UART, 1st Start bit 0, then 8bits data and at last stop bit 1}**

0	D0	D1	D2	D3	D4	D5	D6	D7	1
---	----	----	----	----	----	----	----	----	---

❑ **Mode 2 & 3 {9 Bit UART, 1st Start bit 0, then 8 bits data, 1 bit parity and at last stop bit 1}**

0	D0	D1	D2	D3	D4	D5	D6	D7	P	1
---	----	----	----	----	----	----	----	----	---	---

Example 10-1

With XTAL = 11.0592 MHz, find the TH1 value needed to have the following baud rates. (a) 9600 (b) 2400 (c) 1200

Solution:

With XTAL = 11.0592 MHz, we have:

The machine cycle frequency of the 8051 = $11.0592 \text{ MHz} / 12 = 921.6 \text{ kHz}$, and $921.6 \text{ kHz} / 32 = 28,800 \text{ Hz}$ is the frequency provided by UART to Timer 1 to set baud rate.

- (a) $28,800 / 3 = 9600$ where $-3 = \text{FD (hex)}$ is loaded into TH1
(b) $28,800 / 12 = 2400$ where $-12 = \text{F4 (hex)}$ is loaded into TH1
(c) $28,800 / 24 = 1200$ where $-24 = \text{E8 (hex)}$ is loaded into TH1

Notice that 1/12th of the crystal frequency divided by 32 is the default value upon activation of the 8051 RESET pin. We can change this default setting. This is explained at the end of this chapter.

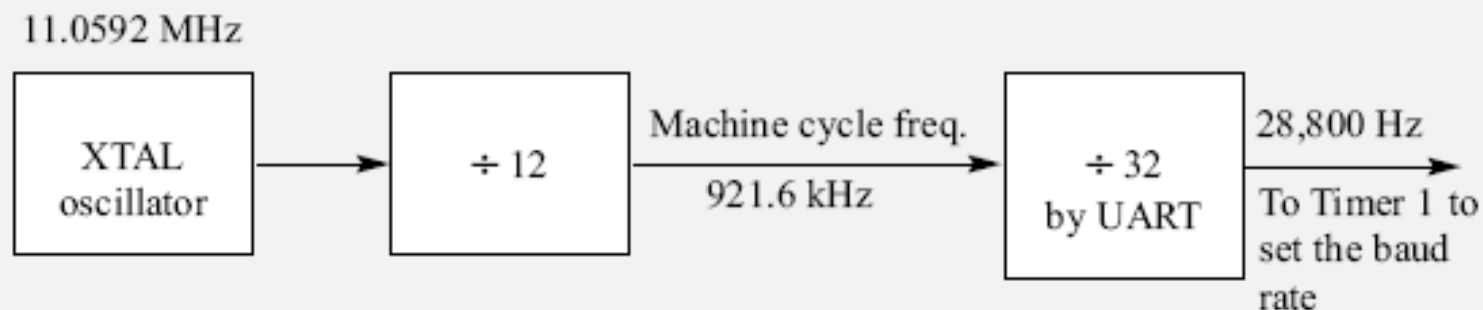


Table 10-4: Timer 1 TH1 Register Values for Various Baud Rates

Baud Rate	TH1 (Decimal)	TH1 (Hex)
9600	-3	FD
4800	-6	FA
2400	-12	F4
1200	-24	E8

Note: XTAL = 11.0592 MHz.

TF is set to 1 every 12 ticks, so it functions as a frequency divider

```
MOV SBUF,#'D'      ;load SBUF=44H, ASCII for 'D'
MOV SBUF,A          ;copy accumulator into SBUF
MOV A,SBUF          ;copy SBUF into accumulator
```


SBUF Register

- The register SBUF is used to hold the data. The special function register SBUF is physically two registers.
- SCON
- PCON

SCON Register (cont.)

- SM0, SM1
 - They determine the framing of data by specifying the number of bits per character, and the start and stop bits

SM0	SM1	
0	0	Serial Mode 0
0	1	Serial Mode 1, 8-bit data, 1 stop bit, 1 start bit
1	0	Serial Mode 2
1	1	Serial Mode 3

Only mode 1 is
of interest to us

- SM2
 - This enables the multiprocessing capability of the 8051

SCON Register (cont.)

- REN (receive enable)
 - When it is high, it allows 8051 to receive data on RxD pin
 - If low, the receiver is disabled
- TI (transmit interrupt)
 - When 8051 finishes the transfer of 8-bit character
 - It raises TI flag to indicate that it is ready to transfer another byte
 - TI bit is raised at the beginning of the stop bit

SCON Register (cont.)

- RI (receive interrupt)
 - When 8051 receives data serially via RxD, it gets rid of the start and stop bits and places the byte in SBUF register
 - It raises the RI flag bit to indicate that a byte has been received and should be picked up before it is lost
 - RI is raised halfway through the stop bit
- SCON is a bit-addressable register

PCON Register- Power Control register

- Register PCON controls processor powerdown, sleep modes and serial data baudrate. Only one bit of PCON is used with respect to serial communication. The seventh bit (b7)(SMOD) is used to generate the baud rate of serial communication.



- It is used to control the 8051MC Power Modes.
- It is located at 87H of the SFR Memory Space.
- Using two bits in the PCON Register, the microcontroller can be set to Idle Mode and Power Down Mode.

SMOD: Serial baud rate modify bit

GF1: General purpose user flag bit 1

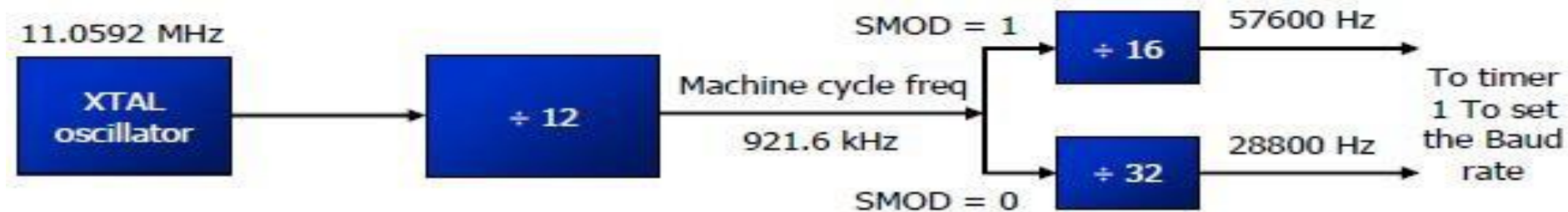
GF0: General purpose user flag bit 0

PD: Power down bit

IDL: Idle mode bit

Bit 7 of PCON is the SMOD bit.

Setting SMOD has the effect of doubling the baud rate in modes 1, 2, and 3. In mode 2, the baud rate can be doubled from a default value of 1/64th the oscillator frequency (SMOD = 0), to 1/32nd the oscillator frequency (SMOD = 1).



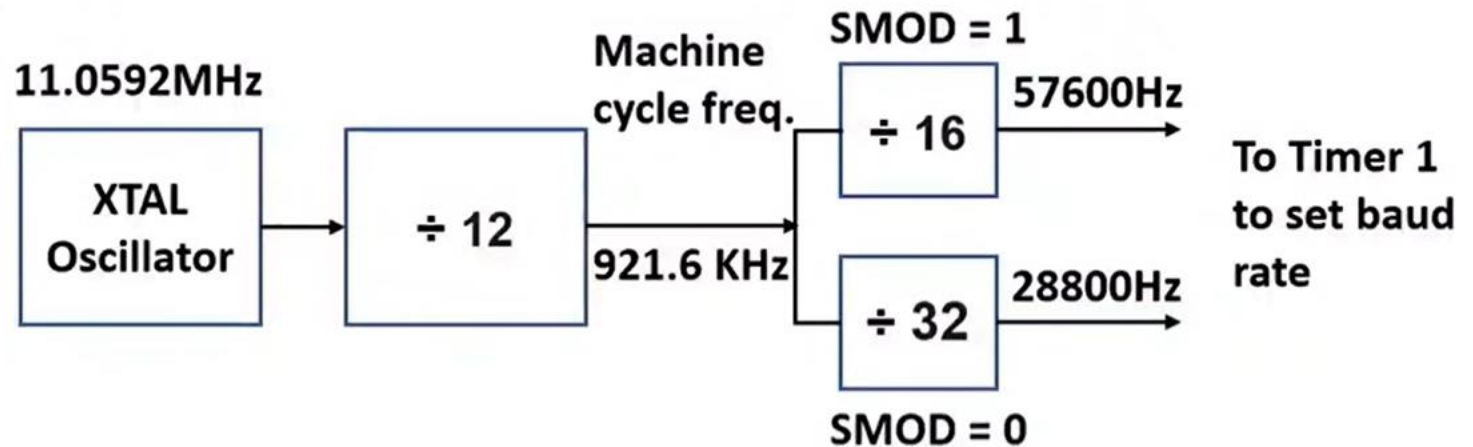
Baud Rate comparison for SMOD=0 and SMOD=1

TH1	(Decimal)	(Hex)	SMOD=0	SMOD=1
-3		FD	9600	19200
-6		FA	4800	9600
-12		F4	2400	4800
-24		E8	1200	2400

❑ The baud rate is the rate at which information is transferred in a communication channel.

❑ In the serial port context, "9600 baud" means that the serial port is capable of transferring a maximum of 9600 bits per second.

110
150
300
600
1200
2400
4800
9600
19200



Standard PC Baud Rate

For 9600 baud rate we will use

TH1 = 0xFD

TMOD = 0x20 for Timer 1 Mode 2

$28,800 / 3 = 9600$ where -3 = FD (hex) is loaded into TH1

$28,800 / 12 = 2400$ where -12 = F4 (hex) is loaded into TH1

$(28,800 / 24 = 1200$ where -24 = E8 (hex) is loaded into TH1

SERIAL TRANSMISSION

Serial TX programming Steps

Step1: TMOD register is loaded with the value 20H, indicating the use of timer 1 in mode 2 (8-bit auto-reload) to set baud rate

Step2: The TH1 is loaded with 0xFD to set baud rate 9600 for serial data transfer

Step3: The SCON register is loaded with the value 50H, indicating serial mode 1, where an 8-bit data is framed with start and stop bits and enable reception

Step4: TR1 is set to 1 to start timer 1

Step5: The character byte to be transferred serially is written into SBUF register

Step6: The TI flag bit is monitored to become 1 to see if the character has been transferred completely

Step7: Clear the TI flag and go to step 5 for transmitting the next byte

SERIAL COMMUNICA- TION PROGRAMMING

Programming Serial Data Transmitting

- ❑ In programming the 8051 to transfer character bytes serially
 1. TMOD register is loaded with the value 20H, indicating the use of timer 1 in mode 2 (8-bit auto-reload) to set baud rate
 2. The TH1 is loaded with one of the values to set baud rate for serial data transfer
 3. The SCON register is loaded with the value 50H, indicating serial mode 1, where an 8-bit data is framed with start and stop bits
 4. TR1 is set to 1 to start timer 1
 5. TI is cleared by `CLR TI` instruction
 6. The character byte to be transferred serially is written into SBUF register
 7. The TI flag bit is monitored with the use of instruction `JNB TI, xx` to see if the character has been transferred completely
 8. To transfer the next byte, go to step 5

SERIAL COMMUNICA- TION PROGRAMMING

Importance of TI Flag

- ❑ The steps that 8051 goes through in transmitting a character via TxD
 1. The byte character to be transmitted is written into the SBUF register
 2. The start bit is transferred
 3. The 8-bit character is transferred on bit at a time
 4. The stop bit is transferred
 - It is during the transfer of the stop bit that 8051 raises the TI flag, indicating that the last character was transmitted
 5. By monitoring the TI flag, we make sure that we are not overloading the SBUF
 - If we write another byte into the SBUF before TI is raised, the untransmitted portion of the previous byte will be lost
 6. After SBUF is loaded with a new byte, the TI flag bit must be forced to 0 by `CLR TI` in order for this new byte to be transferred

SERIAL COMMUNICA- TION PROGRAMMING

Importance of TI Flag (cont')

- ❑ By checking the TI flag bit, we know whether or not the 8051 is ready to transfer another byte
 - It must be noted that TI flag bit is raised by 8051 itself when it finishes data transfer
 - It must be cleared by the programmer with instruction `CLR TI`
 - If we write a byte into SBUF before the TI flag bit is raised, we risk the loss of a portion of the byte being transferred
- ❑ The TI bit can be checked by
 - The instruction `JNB TI, xx`
 - Using an interrupt

Example 10-2

Write a program for the 8051 to transfer letter “A” serially at 4800 baud, continuously.

Solution:

```
                MOV    TMOD,#20H    ;Timer 1, mode 2 (auto-reload)
                MOV    TH1,#-6      ;4800 baud rate
                MOV    SCON,#50H    ;8-bit, 1 stop, REN enabled
                SETB   TR1          ;start Timer 1
AGAIN:          MOV    SBUF,#"A"    ;letter "A" to be transferred
HERE:          JNB     TI,HERE      ;wait for the last bit
                CLR     TI          ;clear TI for next char
                SJMP   AGAIN        ;keep sending A
```

Example 10-3

Write a program to transfer the message "YES" serially at 9600 baud, 8-bit data, 1 stop bit. Do this continuously.

Solution:

```

        MOV    TMOD,#20H ;Timer 1, mode 2
        MOV    TH1,#-3   ;9600 baud
        MOV    SCON,#50H ;8-bit, 1 stop bit, REN enabled
        SETB   TR1       ;start Timer 1
AGAIN:   MOV    A,#"Y"    ;transfer "Y"
        ACALL  TRANS
        MOV    A,#"E"    ;transfer "E"
        ACALL  TRANS
        MOV    A,#"S"    ;transfer "S"
        ACALL  TRANS
        SJMP   AGAIN     ;keep doing it
;-----serial data transfer subroutine
TRANS:   MOV    SBUF,A    ;load SBUF
HERE:    JNB    TI,HERE   ;wait for last bit to transfer
        CLR    TI        ;get ready for next byte
        RET
```

SERIAL RECEPTION

SERIAL COMMUNICA- TION PROGRAMMING

Programming Serial Data Receiving

- ❑ In programming the 8051 to receive character bytes serially
 1. TMOD register is loaded with the value 20H, indicating the use of timer 1 in mode 2 (8-bit auto-reload) to set baud rate
 2. TH1 is loaded to set baud rate
 3. The SCON register is loaded with the value 50H, indicating serial mode 1, where an 8-bit data is framed with start and stop bits
 4. TR1 is set to 1 to start timer 1
 5. RI is cleared by `CLR RI` instruction
 6. The RI flag bit is monitored with the use of instruction `JNB RI, xx` to see if an entire character has been received yet
 7. When RI is raised, SBUF has the byte, its contents are moved into a safe place
 8. To receive the next character, go to step 5

SERIAL COMMUNICA- TION PROGRAMMING

Programming Serial Data Receiving (cont')

Write a program for the 8051 to receive bytes of data serially, and put them in P1, set the baud rate at 4800, 8-bit data, and 1 stop bit

Solution:

```
      MOV  TMOD,#20H    ;timer 1,mode 2(auto reload)
      MOV  TH1,#-6      ;4800 baud rate
      MOV  SCON,#50H    ;8-bit, 1 stop, REN enabled
      SETB TR1          ;start timer 1
HERE:  JNB  RI,HERE      ;wait for char to come in
      MOV  A,SBUF        ;saving incoming byte in A
      MOV  P1,A          ;send to port 1
      CLR  RI            ;get ready to receive next
                        ;byte
      SJMP HERE          ;keep getting data
```